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B. E. (Eighth Semester) Examination, Nov.-Dec., 2015

(Mechanical Engg. Branch)

PRODUCTION MANAGEMENT

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) is compulsory in each question. Attempt any two parts from (b), (c) and (d).*

1. (a) What are the objectives of Production Management? 2

(b) What are the various types of production system? 7

(c) What are the various stages in product design/development process? 7

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(d) An analysis of company reveals the following information : 7

Cost element	Variable Cost	Fixed Cost
Direct Material	32·8	—
Direct Labour	28·4	—
Factory Overheads	12·6	1,89,900
Distribution Overheads	4·1	58,400
General administrative overheads	1·1	66,700
Budgeted sales are	—	18,50,000

Determine :

(i) Break even sales volume

(ii) The profit at budgeted sales volume

(iii) Profit if actual sales drop by 10%

2. (a) What is the role of forecasting in Production Management? 2

(b) The past data regarding the sales of SPMS for last five years is given. Using the least square method, fit a straight line. Estimate the sales for the year 1996 and 1997. 7

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Year	1991	1992	1993	1994	1995
Sales ('00)	35	56	79	80	40

(c) The data given below represents sales figures of ABC company for 12 months of the year 1996. 7

(i) Compute three months moving average (ignoring decimal values)

(ii) Forecast the demand for the month of Jan. 1997

(iii) If actual demand for the month of Jan. 1997 is 905 units, what should be the forecast for the month of Feb. '97?

Month	Sales Rs. (000)	Month	Sales Rs. (000)
Jan.	400	July	710
Feb.	490	August	800
March	570	September	820
April	500	October	910
May	640	November	860
June	680	December	950

(d) Estimate the sales forecast for the year 2000, using exponential smoothing forecaster. Take $\alpha = 0.5$ and the forecast for the year 1995 as 160×10^5 units. 7

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Year	1995	1996	1997	1998	1999
Sales ($X\ 10^5$)	180	168	159	170	188

3. (a) What is scheduling? 2

(b) Explain the various strategies. Which are used in aggregate planning to meet non-uniform demand? 7

(c) Discuss the factors which must be considered for selecting a material handling equipment. 7

(d) There are seven jobs which are to be pressed first on Machine 1 and then on Machine 2. Processing time in hours are given below : 7

Job	A	B	C	D	E	F	G
M/C 1	6	24	30	12	20	22	18
M/C 2	16	20	20	13	24	2	6

Find the optimal sequence and total elapsed time compute the idle time on M/C 2.

4. (a) What are the Objectives of Material Management? 2

(b) What is the importance of store management? What are the functions of stores? What are LIFO and FIFO methods in pricing of stores? 7

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(c) Explain the following methods of inventory classification. 7

Find :

(i) ABC

(ii) VED

(iii) FSN

(d) A manufacturer has to supply his customers 3600 units of his product per year. Shortages are not permitted. Inventory carrying cost amounts Rs. 1·2 per unit per annum. The set-up cost per run is Rs. 80. 7

Find :

(i) Economic Order Quantity

(ii) Otpimum no. of Orders per annum

(iii) Average annual inventory cost (minimum)

(iv) Optimum period of supply per optimum order

(a) What is the difference between variable and attribute control charts? 2

(b) Explain briefly Kanban system of manufacturing. 7

(c) Design and draw OC curve for sample size $n = 10$ and acceptance number $C = 2$. What is AOQL? 7

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(d) Control charts for $\bar{X}$ and R, are maintained on certain dimensions of a manufactured part, measured in mm. The subgroup size is 4. The value of $\bar{X}$ and R are computed for each group. After 20 subgroups $\sum \bar{X} = 412{\cdot}83$ and $\sum R = 3{\cdot}39$. Compute the values of 3 sigma limits for $\bar{X}$ and R charts if the process is in statistical control. 7

Given :

(i) A_2 (Factor for $\bar{X}$ chart) = 0·73

(ii) D_4 (Factor for R chart) = 2·28

(iii) D_3 (Factor for R chart) = 0

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